Markscheme

Mathematics: analysis and approaches
Practice question bank

48. (a)

Consider the product $\log_2 3 \times \log_3 4 \times \log_4 5 \times \log_5 8$.

Find the exact value of this product.

apply the change of base rule to each factor: $\dfrac{\ln 3}{\ln 2} \times \dfrac{\ln 4}{\ln 3} \times \dfrac{\ln 5}{\ln 4} \times \dfrac{\ln 8}{\ln 5}$ *(M1)* **A1**

the intermediate terms cancel, leaving $\dfrac{\ln 8}{\ln 2}$ *(M1)* **A1**

$$= \log_2 8 = 3$$ **A1**

[5 marks]

(b)

Hence evaluate the analogous product $\log_2 3 \times \log_3 4 \times \cdots \times \log_{15} 16$.

by the same telescoping cancellation, this equals $\log_2 16$ *(M1)*

$$= 4$$ **A1**

[2 marks]